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Minimal Example

In this guide we will explain how to install a simple peer-to-peer backup system through the inventory. Such that machines will backup it's state to other machines in the clan, ensuring redundancy and data safety.

What is BorgBackup?

BorgBackup is a powerful and efficient backup solution designed for secure and space-efficient backups. It offers features such as:

- **Deduplication:** Saves storage space by avoiding duplicate data.
- **Encryption:** Ensures backups are secure and authenticated.
- **Compression:** Supports multiple compression algorithms like lz4, zstd, zlib, and more.
- **FUSE Mounting:** Allows backups to be mounted as a file system.
- **Cross-Platform:** Works on Linux, macOS, BSD, and more.
- **Open Source:** Licensed under BSD and supported by an active community.

While this guide uses BorgBackup, you can also use other backup services supported by Clan, depending on your requirements.

Example Setup

In this example, we configure a backup system with three machines: `bob`, `jon`, and `alice`. The `bob` and `jon` machines will periodically back up their state folders to `alice`. The backups are encrypted for security.

```
1 inventory.instances = {  
2     borgbackup = {  
3         module = {
```

```
4     name = "borgbackup";
5     input = "clan-core";
6 };
7     roles.client.machines = {
8         "bob" = {
9             startAt = "*-*-* 04:00:00";
10        };
11        "jon" = { };
12    };
13    roles.server.machines = {
14        "alice" = {
15            address = "alice.example.org";
16        };
17    };
18 };
19 ;
```

Roles

In a Clan Service, roles define how machines participate in the backup system. Each role applies specific Nix configurations to the machine, enabling flexibility and scalability in your backup setup.

- **Client:** These machines create backups and send them to designated destinations. Clients can be configured to back up to multiple destinations, ensuring redundancy and reliability.
- **Server:** These machines act as repositories, receiving and securely storing backups from client machines. Servers can be dedicated backup nodes within your clan network, providing centralized storage for all backups.

Backup destinations

This service allows you to perform backups to multiple `destinations`. Destinations can be:

- **Local:** Local disk storage
- **Server:** Your own borgbackup server (using the `server` role)
- **Third-party services:** Such as Hetzner's Storage Box

However, if BorgBackup does not meet your needs, you can implement your own backup clan service.

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Digging Deeper

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